

Replogle Miscalibration Exploration

Gene

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This analysis explores our miscalibration issues with Replogle rd7 by comparing to Velten's analysis.

Sceptre Object Comparison

Let's first compare our `sceptre_object` to Velten's:

```
crispat_sceptre_object
```

```
## An object of class sceptre_object.  
##  
## Attributes of the data:  
##   • 17857 cells (17857 after cellwise QC)  
##   • 36601 responses  
##   • Low multiplicity-of-infection  
##   • 39 targeting gRNAs (distributed across 39 targets)  
##   • 113 non-targeting gRNAs  
##   • 11 covariates (Batch, cell, G2M_score, grna_n_nonzero, grna_n_umis, percent_mito, response_n_nonzero)
```

```
our_sceptre_object
```

```
## An object of class sceptre_object.  
##  
## Attributes of the data:  
##   • 616184 cells  
##   • 36601 responses  
##   • Low multiplicity-of-infection  
##   • 2536 targeting gRNAs (distributed across 2384 targets)  
##   • 130 non-targeting gRNAs  
##   • 6 covariates (batch, grna_n_nonzero, grna_n_umis, response_n_nonzero, response_n_umis, response_p)
```

The main differences are that

1. they have way fewer cells,
2. they have 17 fewer NT gRNAs, and
3. they have extra covariates `G2M_score` and `S_score`, which encode cell cycle state.

The first is largely irrelevant to us because what we care about are the cells with NT gRNAs only, and so we don't care about the other cells anyways. An inspection of Velten's codebase revealed that they retained all cells with NT gRNAs, *except* those containing 17 NT gRNAs found by Replogle to have apparent effects on genes. This brings us to difference 2, which is very important. Throwing out NT gRNAs with an apparent effect on genes prior to a calibration check is circular reasoning, so we would want to repeat the calibration check with all NT gRNAs. Item 3 may be important; if the cell cycle state acts as a confounder. Let's check whether this is the case in the next section.

Cell Cycle Effects on Non-targeting gRNAs

Cell cycle state is known to impact gene expression; let's check based on Velten's data whether it also impacts which gRNA a cell received.

```
aov(S_score ~ grna, data = velten_df) |> summary()
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## grna          112     5.4  0.04800   1.486 0.000695 ***
## Residuals    12611   407.4  0.03231
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

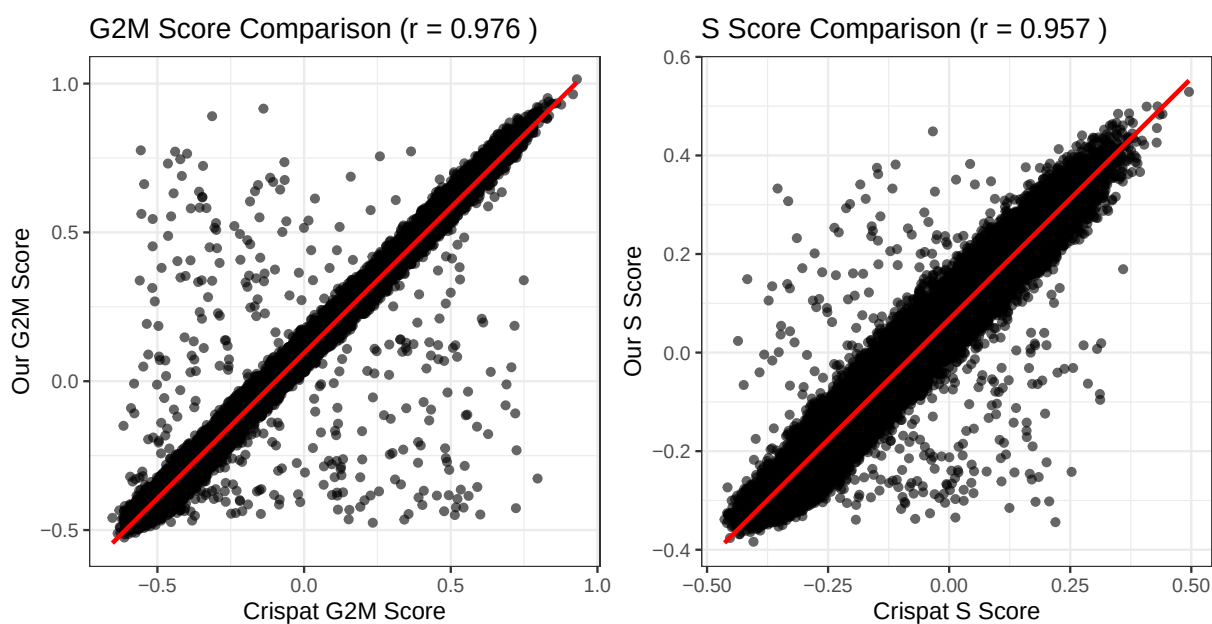
```
aov(G2M_score ~ grna, data = velten_df) |> summary()
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## grna          112    22.1  0.1976   1.452 0.00134 **
## Residuals    12611  1716.2  0.1361
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

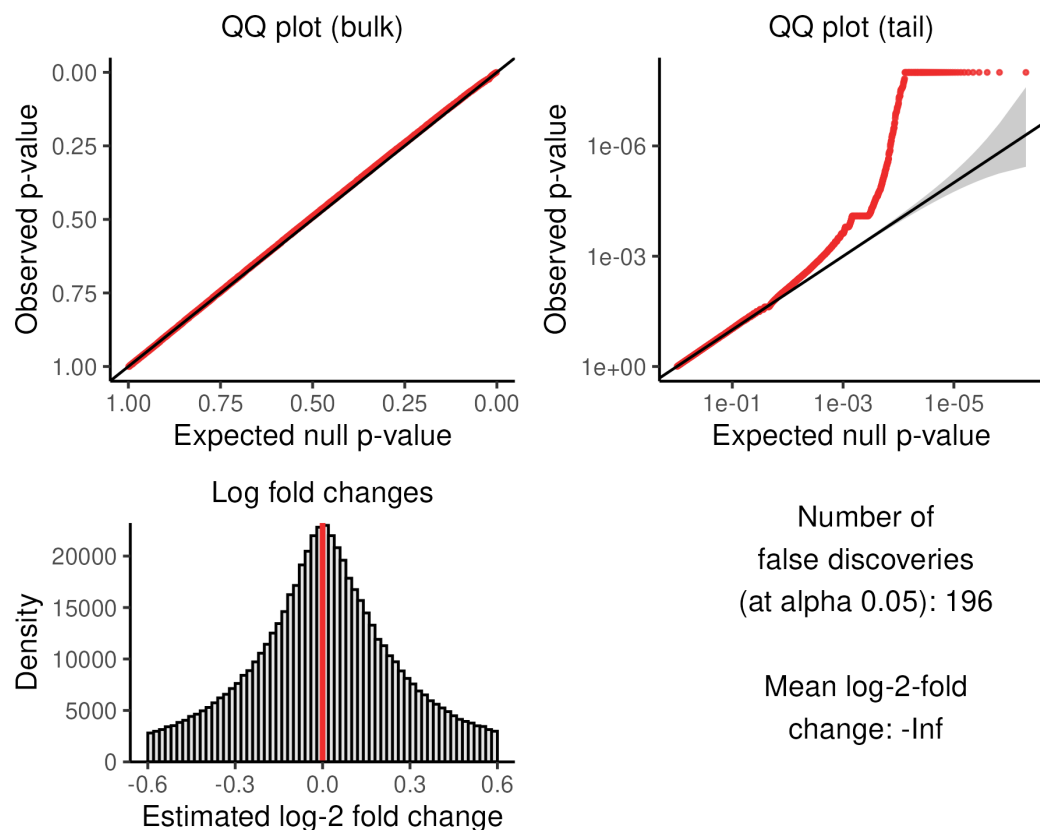
We see that both cell cycle variables are correlated with gRNA, which makes cell cycle a possible confounder. Maybe controlling for cell cycle is enough to explain the miscalibration we're seeing.

Calibration Check with Cell Cycle Scores

We can't rely on the Velten data to get the cell cycle scores for our calibration check, since they excluded 17 gRNAs. Therefore, I added functionality to `ondisc` to calculate cell cycle during import (which I hope will be independently useful). Below is a check that the cell cycle scores I computed from scratch match those from the Velten data, for the cells that overlap.



The agreement seems pretty good. I then reran the calibration check using this newly imported data on a million negative control pairs, with the same formula (and other analysis parameters) used by Velten.



Unfortunately, the miscalibration remains. Below is a table of how the false positives are distributed across gRNAs:

```
## # A tibble: 49 x 2
##   grna_target      n
##   <fct>          <int>
## 1 non-targeting_vector_87    49
## 2 non-targeting_vector_59    21
## 3 non-targeting_vector_55    17
## 4 non-targeting_vector_63    14
## 5 non-targeting_vector_1      8
## 6 non-targeting_vector_112    8
## 7 non-targeting_vector_71     8
## 8 non-targeting_vector_34     6
## 9 non-targeting_vector_75     6
## 10 non-targeting_vector_106    4
## 11 non-targeting_vector_117    3
## 12 non-targeting_vector_13     2
## 13 non-targeting_vector_130    2
## 14 non-targeting_vector_16     2
## 15 non-targeting_vector_2      2
## 16 non-targeting_vector_28     2
## 17 non-targeting_vector_44     2
## 18 non-targeting_vector_47     2
## 19 non-targeting_vector_49     2
## 20 non-targeting_vector_61     2
## # i 29 more rows
```

So we have about 20 NT gRNAs that make up the bulk of the problem; likely these contain the 17 filtered out by Replogle and Velten.

Conclusion

Adding cell cycle didn't resolve the calibration issue, leading me to believe that the Replogle data simply have NT gRNAs that “didn't work”, in the sense that they do have effects on some genes. In my opinion, this means that the Replogle data are too low-quality for us to analyze.